



Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering & Research
IQAC PCCOER



Academic Year: 2022-23

Faculty Profile



Name of Faculty:	Dr. Sham H. Mankar		
Qualification :	Ph.D. (Mech. Engg.), M.E. (Mech.)		
Designation :	Associate Professor		
Department :	Mechanical Engineering		
Date of Joining :	13 June 2017		
Experience :	Teaching : 16.2 Years Industry: 4.6 Years Research: 03 Years		
Research Areas / Area of Specialization	Advanced Manufacturing Processes		
Courses / Subjects Taken:	Manufacturing Processes- I, Manufacturing Processes- II, Advanced Manufacturing Processes, Automobile Engineering, Industrial Engineering, Production Technology, Industrial Engineering and Technology Management, Robotics.		
IPR :	Title	Docket No./ Diary No.	Registered /Filed
	Patent on Mechanical properties enhancement in Gas Metal Arc welded joints of Aluminium Alloys using Copper base plates.	201821014963	Under Examination
	Copyright on Research Planning for Parametric Studies of Aluminium Alloy Weld Joint Using GMAW Process.	16822/2017-CO/L, Date: 29/11/2017.	Registered
Research Publications :	09		
Journal Publications:	Year	Title of Paper	Published in
	June 2020	Heat and Mass Transfer Analysis of Solid Desiccant Dehumidifier Wheel, International Journal of Mechanical and Production Engineering Research and Development	IJMPERD, ISSN (P): 2249-6890, ISSN (E): 2249-8001, Vol. 10, Issue 3, 9597-9612.

	August 2020	Comparative Analysis of Different sections of Evaporator Tubes in Domestic Refrigerator,	Waffen-Und Kostumkunde Journal ISSN: 0042-9945, Vol. XI, Issue VIII, 211-218.
	September 2020	Experimental, Design and CFD simulation to obtain optimum inclination angle for solar still	Waffen-Und Kostumkunde Journal ISSN: 0042-9945, Vol. XI, Issue IX, 1-6.
	August 2018	Prediction of Bead Geometry Parameters using Response Surface Methodology and Finite Element Analysis of Gas Metal Arc Welded Joints of Aluminum Alloy	Journal of Experimental & Applied Mechanics
	2017	Topology optimization and analysis for double acting compliant retractor mechanism	Int J Curr Eng Technol
	June 2016	Experimental investigation on the effects of process parameters of GMAW and transient thermal analysis of AA7075 T6	International Engineering Research Journal (IERJ), ISSN 2395-1621, 1109-1118.
	2016	Numerical and experimental stress analysis of a composite leaf spring	International Journal of Engineering and Technology (IJET). ISSN (Print: 2319-8613 ISSN (Online): 0975-402
	2016	Vibration analysis of a small wind turbine blade	International Journal of Engineering and Technology (IJET)
	Nov. 2015	Design and Experimental Analysis of Aluminium Alloy Weld Joint by Using Pulse - GMAW Process	Journal (IERJ),ISSN 2395-1621, Special Issue 2, Nov. 2015, 4012-4016.
	Aug.-Sep. 2015	Design and Experimental Analysis of Aluminium Alloy Weld Joint by Using GMAW Process,	International Journal of Emerging Trends in Engineering and Development, ISSN 2249-6149, Vol. 5 (Aug.-Sep. 2015), Issue 5, 43-52.
	2015	A Review on Design and Optimization of Composite Leaf Spring,	International Journal of Innovative and Emerging Research in Engineering (e-ISSN: 2394 - 3343, p-ISSN:

			2394 – 5494), Volume 2, Sp. Issue 1, MEPCON 2015,187-190
	2015	Computational and Experimental Static Stress Analysis of Composite Leaf Spring for Light Motor Vehicle	International Journal of Applied Engineering Research
	2011	Stability and ergonomic design features of tractor mounted hydraulic elevator for coconut harvesting	American Journal of Engineering and applied sciences
Conference Publications:	Year	Title of Paper	Published in
	2017	Weight and Cost Reduction of A Small Wind Turbine Blade	IEEE Sponsored 2nd International Conference for Convergence in Technology (I2CT) – 2017, 7 – 9 Aril 2017, Pune
	2016	Topology Optimization Methods for Compliant Mechanisms	International Conference on Design, Manufacturing and Mechatronics ICDMM – 2016, 19-20 Feb. 2016. T.C.O.E.R., Pune
	2016	Study of Inter-laminar Shear Stress in Composite Structure	International Conference on Design, Manufacturing and Mechatronics ICDMM – 2016, 19-20 Feb. 2016. T.C.O.E.R., Pune.
	2015	Experimental Analysis of GMAW welded joint of AA7020 using Finite Element Analysis	International Conference on Design, Manufacturing and Mechatronics ICDMM – 2015, 11-13 Feb. 2015. K.J.C.O.E and M. R., Pune.
	2015	Design and Optimization of Composite Leaf spring for Light Motor Vehicle	International Conference on Design, Manufacturing and Mechatronics ICDMM – 2015, 11-13 Feb. 2015. K.J.C.O.E and M. R., Pune.
	2015	Design, Analysis and Optimization Electric Overhead Travelling Crane Hook	International Conference on Design, Manufacturing and Mechatronics ICDMM – 2015, 11-13 Feb. 2015. K.J.C.O.E and M. R., Pune.

	2014	Surface Creation Using Scanned Point Clouds with Sharp Features	International Conference ISET 2014 at T.C.O.E.R. Pune, 09-10 Jan. 2014.
	2014	Influence of Welding Parameters on the Microstructure and Mechanical Properties of Multipass Submerged Arc Welding	International Conference ISET 2014 at T.C.O.E.R. Pune, 09-10 Jan. 2014.
	2014	Analysis of Fluid and Thermal Flow due to Disproportional Heating	International Conference ISET 2014 at T.C.O.E.R. Pune, 09-10 Jan. 2014.
	2014	Mechanical and Structural Characterization of GMAW AA7020 Multipass Butt Weld Joint	International Conference ISET 2014 at T.C.O.E.R. Pune, 09-10 Jan. 2014.
	2011	Process Parameter Optimization of Aluminum Alloy 6061 with Pulsed Gas Tungsten Arc Welding (GTAW)	International Conference at Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat (Gujarat), (ICAME-2011), 06-08 June 2011.
	2011	Recent Advances of Arc Welding Improves the Efficiency of Farm Machines	National Conference at Pimpri Chinchwad College of Engineering, Nigdi, Pune (M.S.) (NCAEMS-2011), 28-29 April 2011.
	2006	Parabolic Solar Concentrator using Digital Tracking	International Conference at Adhiyamaan College of Engineering (ISTE Chapter-2006), Hosur (Tamilnadu), 06-08 Sept.2006.
Book's Publication:	NIL		
Profile Link :	Google Scholar Profile: https://scholar.google.com/citations?view_op=list_works&hl=en&user=5yp7xiEAAAAJ		
H -Index:	03		
Product Developed:	NIL		
Certifications :	Internal ISO Auditor		
Awards/Appreciation/Achievements:	Successfully Completed NPTEL course on Teaching And Learning in Engineering (TALE)		

Membership:	Life Member of ISTE and QCFI
Industrial Training :	Ascent Casting Technology Chakan, Pune. 09/06/2018 to 15/06/2018
Resource Person /Expert Session/Jury Invited:	Worked as Ph.D. Examiner for two students from Sant Gadge Baba Amravati University.
Workshop/FDP/STTP attended (more than a Week):	1. One Week Faculty Development Programme on 'Future Materials: Nanocomposites' from 15th June, 2020 to 21st June, 2020 organized by Department of Mechanical Engineering. Bharti Vidyapeeth (deemed to be university), College of Engineering, Pune. 2. One Week Faculty Development Programme on 'Research Opportunities in Advanced Manufacturing Processes' from 22nd June, 2020 to 28th June, 2020 organized by Department of Mechanical Engineering of BVCOE, Pune. 3. NPTEL Course on Teaching And Learning in Engineering (TALE) 4. One week online Short Term Training Program on "Innovative & Inventive Problem Solving" organised by Department of Production Engineering, AISSMS COE, Pune during 22nd June to 27th June 2021. 5. One Week National workshop on "Human Values and Awareness during COVID -19" during 5 July to 9 July 2021, Organized by Pimpri Chinchwad College of Engg, Akurdi, Pune. 6. One week online Faculty Development Program on "Application Oriented Advanced Functional Nanocomposites (AOAFNC-2021)" during July 26-30, 2021 Organized by Sinhgad College of Engineering, Pune. 7. Five Days Intensive Workshop on "Manuscript preparation for International Journal" during July 05-09, 2021 Organized by SIG Coordination Team, Pimpri Chinchwad College of Engg, Akurdi, Pune.
Event Organized Inter department/Inter college/ Other recognized Industry/IIT's/NPTEL/ SPOKE N TUTORIAL/ etc. :	1. Syllabus Drafting and detailing for SPPU BE Mech. 2015 Course. 2. Syllabus implementation for SPPU BE Mech. 2015 Course. 3. Syllabus implementation for SPPU TE Mech. 2019 Course.
Event Organized in Department:	1. Organized Workshop on "Soft skills" for students 2. Organized "TECHNOVAT'20" under MESA 3. Organized Hands on Workshop on "DESIGN OF ELECTRIC VEHICLE" 4. Organized Hands on Workshop on "Android Based Automation" 5. CIPCIS 2019 and CIPCIS 2020 6. Faculty Development Program on "Emerging Trends in Refrigeration & Air Conditioning" in association with ISHRAE and Institution of Engineers (INDIA) Pune.
University level appointments:	7. Worked as SPPU Subject Chairman and paper Setter for SE (Mech.). 8. Worked as SPPU Subject Chairman and paper Setter for BE (Mech.) 9. Worked as Course Coordinator of Team for Revision, Detailing and Implementation of SPPU TE Mech. Engg. 2019 Course Curriculum for ELECTIVE - 1 Advanced Forming & Joining Processes (302045-A). 10. Worked as Member of Team for Revision, Detailing and Implementation of SPPU TE Mech. Engg. 2019 Course Curriculum for ELECTIVE - 1 Machining Science and Technology (302045-B).

	11. Worked as Member of Team for Revision, Detailing and Implementation of SPPU TE Mech. Engg. 2019 Course Curriculum for Digital Manufacturing Laboratory (302046).
Administrative Duties:	1. Worked as Head of Mechanical Engg. Department from 01/08/2017 to 27/05/2021. 2. Working as IQAC Coordinator 3. Working as IIC President 4. Working as Coordinator for AICTE- CII Survey
Academics Duties:	1. Working as Higher Studies and Competitive Exams Coordinator. 2. Working as Industry Institute Interaction Cell Coordinator 3. Department NBA Coordinator 4. NBA Criterion 1 Coordinator